THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 10

Full Syllabus (Chapters 1–13) | Class 9 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** a broad idea that helps make sense of many situations	principles are broad ideas
2	**(a)** 10 000 litres	0.5 × 20 000
3	**(d)** 4 %	$\frac{2}{50} \times 100 = 4$
4	**(b)** water	the substance present in larger amount that dissolves the solute
5	**(d)** 12 g	$\frac{4}{3} \times 9 = 12$
6	**(a)** 3	N≡N triple bond (5 valence electrons each)
7	**(d)** iodine-131	radioactive iodine
8	**(a)** 15 m s^{-1}	1500 ÷ 100
9	**(b)** 9.8 m s^{-2}	g
10	**(c)** keeps moving indefinitely if no friction acts	basis of the first law
11	**(a)** force of friction	reaction to backward push
12	**(c)** 1000 W	kilo = 10^3
13	**(d)** compression	particles crowded
14	**(c)** leucoplasts	storage
15	**(c)** DNA and ribosomes	evolutionary history
16	**(b)** hinge joint	movement in one plane
17	**(a)** flatworms	Platyhelminthes
18	**(d)** HIV/AIDS	use of condoms prevents it
19	**(c)** IVF	test-tube baby
20	**(a)** transpiration	stomata

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	A reasoned expectation about what will happen based on laws, theories, models and evidence (1); a guess has no such basis (1).	1+1
22	In a compound elements are chemically combined in fixed ratio; in a mixture the components can be mixed in any ratio (2).	2
23	Displacement divided by the time interval (1); m s^{-1} (1).	1+1
24	An object at rest remains at rest and an object in motion continues with constant velocity unless a net force acts on it (2).	2
25	The diffusion of water across a selectively permeable membrane from a dilute to a concentrated solution (2).	2
26	Tissues of actively dividing cells (1); at the growing tips and along the circumference of stems (1).	1+1
27	New plants arise from vegetative parts (1); potato tuber and Bryophyllum leaf (1).	1+1
28	Solar constant: energy at the top of the atmosphere (about 1.4 kW m^{-2}); insolation: radiation reaching the surface (1+1).	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	The mixture stands to form two layers (1); the lower layer is run out through the stopcock, the interface portion is collected separately, then the upper layer (1); based on immiscibility and difference in density (1).	1+1+1
30	Na (2,8,1) loses 1 electron, valency 1 (1); O (2,6) gains 2, valency 2 (1); C (2,4) shares 4, valency 4 (1).	1+1+1
31	Effort is smaller because it is applied over a longer distance (2); work done (ignoring friction) is the same (1).	2+1
32	Distance = 2.4 m (1); t = 2.4/345 (1) ≈ 0.007 s (1).	1+1+1

33	Voluntary: skeletal, e.g. leg muscles; involuntary: smooth and cardiac, e.g. intestine and heart (1.5+1.5).	1½+1½
34	Hard exoskeleton, jointed appendages, segmented body, adaptation to land, water and air (3).	3
35	Total reactants $= 6.92$ g (1); mass of products must equal this (1); $CaCl_2 = 6.92 - (1.76 + 0.72) = 4.44$ g (1).	1+1+1
	OR: Na (2,8,1) loses 1 electron to become Na^+ (2,8) (1); Cl (2,8,7) gains 1 electron to become Cl^- (2,8,8) (1); opposite charges attract forming an ionic bond (1).	1+1+1
36	Less transpiration and less infiltration, more runoff and erosion (1.5); less rainfall and soil loss (1.5).	1½+1½
	OR: Save energy, plant trees, use renewable energy, conserve water, sustainable farming (1+1+1).	1+1+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) $CaCl_2$, Al_2O_3, Na_2SO_4, MgO, NH_3 (2.5). (ii) Cross-over of valencies / criss-cross method (1). (iii) carbon monoxide, carbon dioxide, carbon disulfide, dinitrogen tetroxide, phosphorus trichloride (1.5).	2½+1+1½
	OR: (i) A neutral entity of more than one atom that can exist independently (1). (ii) Element: H_2, O_2, Cl_2; compound: H_2O, HCl, CO_2 (3). (iii) Its valence shell is already full (1).	1+3+1
38	(i) Weight downwards, her upward force (1). (ii) 294 N (1). (iii) 294 N upwards (1). (iv) Zero (2).	1+1+1+2
	OR: (i) Treat both and the string as one object; internal tension is ignored, $a = F/(m_1 + m_2)$ (2). (ii) 2 m s^{-2} (1.5). (iii) $T = m_2 a = 6$ N when box 1 is pulled (1.5).	2+1½+1½
39	(i) They contain actively dividing cells (1). (ii) Cell division (mitosis) (1). (iii) Mitosis and meiosis (1). (iv) Growth, repair, replacement and asexual reproduction (2).	1+1+1+2
	OR: (i) Orderly sequence of growth and division of a cell (1). (ii) Number of divisions, daughter cells, chromosome number, purpose (3). (iii) Growth, repair, reproduction (1).	1+3+1
40	(i) Testes, sperm duct, urethra; ovaries, oviducts, uterus, vagina (3). (ii) Testes: sperms and hormones; ovaries: eggs and hormones (2).	3+2
	OR: (i) Puberty (1). (ii) Hormones (1). (iii) Onset of menstruation, development of breasts (1). (iv) Supports rapid growth (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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